ROOT CAUSE UNLOCKED

Unlock Your GUT

A Complete Functional Medicine Guide to Healing Intestinal Hyperpermeability and Ending the Chronic Inflammation Cycle

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BEFORE YOU START

This ebook is written for educational purposes only and does not constitute medical advice, diagnosis, or treatment. The information reflects a functional medicine perspective focused on root cause identification. Always work with a licensed provider before changing any supplement or dietary protocol. This is the same clinical framework used with patients at Root Health daily.

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INTRODUCTION

The Gut Is Where Everything Starts

There is a pattern that repeats itself constantly in functional medicine practice. A patient arrives after seeing multiple specialists. They have been to gastroenterology, to rheumatology, to endocrinology, sometimes to psychiatry. They carry a folder of test results, all of which are reported as normal or within range. And yet they are exhausted in a way that sleep does not fix. Their thinking is slow and unreliable. Their skin breaks out in patterns no dermatologist has been able to explain. Their hormones fluctuate in ways that do not respond to the usual interventions. They gain weight despite reasonable eating. They feel anxious in a way that does not correspond to the actual circumstances of their life.

In a significant portion of these patients, the answer is in the gut. Not because the gut is the only problem, but because the gut is the entry point for a cascade of dysfunction that eventually manifests everywhere else. A compromised intestinal barrier allows substances into the bloodstream that do not belong there. The immune system mounts a response. That response, when it becomes chronic, drives systemic inflammation that affects every organ system in the body. The fatigue, the brain fog, the hormonal dysregulation, the skin conditions, the mood changes. They are downstream consequences of a gut barrier that has lost its integrity, not separate problems.

This is the clinical reality of intestinal permeability, commonly called leaky gut. It is not a fringe concept invented by alternative medicine practitioners. The tight junction proteins that regulate gut barrier function (occludin, claudin, and zonulin) have been the subject of hundreds of peer-reviewed studies. The scientist who discovered the zonulin protein, Dr. Alessio Fasano at Harvard Medical School, has published extensively on the central role of intestinal permeability in autoimmune disease, metabolic dysfunction, and neurological disorders.[1] The research is there. The clinical application of that research is what most patients have never had access to.

This guide is that access. What follows is a complete framework for understanding how the gut barrier breaks down, what specific mechanisms are responsible, how to test for it accurately, and what a systematic recovery protocol looks like: beginning with the drainage foundations that must come first, through dietary intervention, targeted supplementation, and the often-overlooked connections between gut health and thyroid autoimmunity that can change the trajectory of a patient's entire health picture.

The goal is to give you a clinical understanding of what is actually happening in your body and why the approach in this guide works when others have not, rather than a list of supplements to buy.

KEY 1

What Leaky Gut Actually Is

The intestinal lining is one of the most remarkable structures in the human body. Stretched flat, the surface area of the small intestine alone covers an area roughly the size of a studio apartment, about 30 to 40 square meters,[2] compressed into a tube less than an inch in diameter through a system of microscopic finger-like projections called villi, and even smaller hair-like structures called microvilli that together form what is known as the brush border. This extraordinary surface area exists for one purpose: to maximize the absorption of nutrients from digested food.

But absorption cannot be indiscriminate. The same surface that absorbs nutrients must also serve as a barrier against everything in the gut lumen that should not enter systemic circulation: undigested food particles, bacterial toxins called lipopolysaccharides (LPS), fungal metabolites, parasites, and the contents of dead and lysed bacteria. The mechanism that performs this selective gatekeeping is a system of protein complexes called tight junctions.

Far from passive seals, tight junctions are dynamic structures that respond to signals from the gut environment, opening and closing in response to nutrients, pathogens, inflammatory signals, and a critical regulatory protein called zonulin. When the gut is healthy and the tight junction proteins (occludin, claudin-1, claudin-3, and ZO-1) are intact and properly assembled, the barrier functions correctly. Only properly digested nutrients cross into the bloodstream. Everything else remains in the gut lumen until it is excreted.

Intestinal permeability occurs when this tight junction system is disrupted. The gaps between intestinal cells widen. Substances that normally would not have access to the bloodstream, including partially digested proteins, bacterial endotoxins, fungal toxins, and lipopolysaccharides from the outer membranes of gram-negative bacteria, cross into systemic circulation. The immune system, which is heavily concentrated in the gut-associated lymphoid tissue (GALT) surrounding the intestinal lining, detects these foreign substances and mounts a response.

In the short term, this immune response is appropriate and protective. The problem emerges when the gut barrier remains compromised. If the tight junction disruption is not resolved, the immune activation does not resolve either. It becomes chronic. And chronic, low-grade immune activation in response to a continuously leaking gut barrier is the mechanism that drives the systemic consequences that patients experience as seemingly unrelated conditions: fatigue, brain fog, skin eruptions, hormonal instability, autoimmune activation, and metabolic dysfunction.

Why This Is Bigger Than the Gut

The gut is far more than a digestive organ that happens to sometimes malfunction. It is arguably the most complex immunological and neurological structure in the body outside the central nervous system. Approximately 70 percent of the body's immune tissue is concentrated in the mucosa-associated lymphoid tissue of the gut. The enteric nervous system, sometimes called the second brain, contains over 100 million neurons, more than the spinal cord, and communicates bidirectionally with the brain via the vagus nerve. The gut produces more than 90 percent of the body's serotonin and significant quantities of dopamine precursors, GABA, and other neuroactive compounds.

This architecture means that what happens in the gut does not stay in the gut. When the gut barrier is chronically compromised, the consequences are systemic. LPS endotoxins crossing the gut barrier activate toll-like receptor 4 (TLR4) in immune cells throughout the body, triggering a pro-inflammatory cytokine cascade that affects the liver, the brain, the thyroid, the adrenals, and the cardiovascular system. Neuroinflammation from gut-derived immune activation has been directly linked in the research literature to depression, anxiety, cognitive impairment, and neurodegenerative disease. Thyroid dysfunction, hormonal dysregulation, and autoimmune activation are all documented downstream consequences of chronic intestinal permeability.

This is the clinical framework that explains why patients with leaky gut present with such diverse and apparently disconnected symptom pictures. The gut is not one of ten problems. In many cases, it is the upstream driver of all the others.

The Diagnostic Gap in Conventional Medicine

Conventional gastroenterology focuses primarily on structural pathology: polyps, ulcers, tumors, masses, and the inflammatory bowel diseases that produce visible changes on colonoscopy or endoscopy. These are important conditions. But they represent only a fraction of the gut dysfunction that drives chronic illness in the broader population. A colonoscopy can be completely normal in a patient with profound intestinal permeability, significant dysbiosis, severely depleted secretory IgA, active H. pylori infection, and measurable Candida overgrowth. Every one of these findings would be invisible on a standard gastroenterological workup.

Intestinal permeability is a functional condition. It requires functional testing to identify. The zonulin antibody, the occludin/actomyosin antibodies, the tight junction protein markers, the secretory IgA measurement, and the fecal gliadin: none of these are part of routine gastroenterological or primary care practice. They have to be specifically ordered, and most physicians have not been trained in their interpretation.

This is a curriculum gap, not a criticism of conventional medicine practitioners. The gap leaves patients circulating through multiple specialties, collecting normal test results, while the actual driver of their symptoms goes undetected. The purpose of this guide is to close that gap: to give you the testing framework and the treatment roadmap that addresses what conventional testing cannot find.

KEY 2

What Breaks the Gut Barrier

Intestinal permeability rarely develops from a single cause in isolation. In the patients who are most symptomatic and most resistant to recovery, multiple factors have converged over time to erode the gut barrier from several directions simultaneously. Understanding the specific mechanisms by which the barrier breaks down directly determines which interventions will be most effective for a given patient.

Gliadin, Gluten, and the Zonulin Response

Of all the dietary factors that affect intestinal permeability, the relationship between gliadin, the immunogenic protein fraction of gluten found in wheat, rye, and barley, and the zonulin signaling pathway is the best documented and arguably the most clinically significant. Gliadin directly triggers the release of zonulin from intestinal epithelial cells, and zonulin is the primary physiological regulator of tight junction opening. When zonulin rises, tight junctions open. The mechanism has been demonstrated in vitro, in animal models, and in human clinical research by Dr. Fasano's group and numerous independent investigators.

What makes this particularly important is that the zonulin response to gliadin is not exclusive to patients with celiac disease. Research has demonstrated that gliadin activates the zonulin pathway in all individuals regardless of celiac status, though the magnitude and duration of the response vary substantially based on genetics, microbiome composition, and baseline gut health.[3] For patients with HLA-DQ2 or DQ8 variants, or with pre-existing gut compromise, the zonulin response to gliadin exposures can produce measurable tight junction disruption that persists for hours after consumption.

The clinical implication is significant: for patients with confirmed or suspected intestinal permeability, gluten reduction is not sufficient. Complete elimination is the therapeutic standard, at least through the active repair phase, because even small and inconsistent exposures maintain the zonulin signaling that keeps tight junctions chronically disrupted. This is why patients who "mostly" avoid gluten do not experience the same degree of improvement as those who eliminate it completely.

Dysbiosis: When the Ecosystem Collapses

The human gut microbiome contains approximately 38 trillion microorganisms representing thousands of species.[4] The microbiome actively participates in the maintenance of gut barrier integrity through multiple mechanisms: producing short-chain fatty acids that serve as the primary fuel source for colonocytes, regulating tight junction protein expression,[5] stimulating secretory IgA production, competing with pathogens for nutrients and binding sites, and modulating the immune tone of the gut mucosa.

When this ecosystem is disrupted through antibiotic use, dietary changes, chronic stress, toxin exposure, or pathogenic infection, the consequences for gut barrier integrity can be profound. Gram-negative bacteria that overgrow in dysbiotic states produce lipopolysaccharides (LPS) as a component of their outer membrane.

When these bacteria die or are lysed, LPS is released. In a healthy gut with intact barrier function, LPS remains in the gut lumen and is excreted. In a dysbiotic gut with compromised barrier function, LPS crosses into circulation and activates the systemic inflammatory cascade through TLR4 receptors on immune cells throughout the body. This is the mechanism of what researchers call “metabolic endotoxemia,” a chronically elevated circulating LPS burden that drives low-grade systemic inflammation even in the absence of any acute infection.[6]

Candida albicans presents a different but equally important mechanism. In its commensal yeast form, Candida is a normal constituent of the gut microbiome. When conditions shift toward high sugar intake, antibiotic use, immune suppression, or prolonged gut inflammation, Candida can transition to its hyphal form, in which it produces thread-like projections that physically penetrate between intestinal epithelial cells. These hyphae mechanically disrupt the tight junction architecture, creating gaps in the barrier that allow other substances to cross into circulation. Candida overgrowth also produces acetaldehyde and other metabolic toxins that damage the gut lining directly.

H. pylori, which colonizes the stomach and upper small intestine in a significant percentage of the population, affects gut barrier integrity through a different mechanism. H. pylori degrades the protective mucus layer that shields the stomach and duodenal epithelium from acid and microbial insult. Without this mucus barrier, the epithelium is exposed to chronic low-grade damage that impairs the integrity of the entire upper gut lining and predisposes to downstream intestinal permeability.

The Role of NSAIDs, Antibiotics, and Medications

Few pharmaceuticals contribute to gut barrier disruption as consistently in clinical practice as non-steroidal anti-inflammatory drugs. NSAIDs inhibit cyclooxygenase enzymes (COX-1 and COX-2), which are responsible for producing prostaglandins throughout the body. What is often overlooked is that prostaglandins in the gut mucosa serve a protective function: they stimulate mucus secretion, regulate blood flow to the intestinal lining, and support epithelial cell turnover. When COX enzymes are inhibited, this prostaglandin-dependent mucosal protection is reduced, and the gut lining becomes more vulnerable to damage from acid, bile, and microbial products.

The gut-damaging effects of NSAIDs are well-established in the research literature. Studies using capsule endoscopy have documented mucosal erosions, ulcerations, and barrier disruption in the small intestine of patients taking NSAIDs, findings that are invisible on standard endoscopy because they occur in segments of the gut that endoscopes cannot reach.[7] Patients who rely on ibuprofen, aspirin, or naproxen regularly for pain management are often maintaining a state of chronic gut barrier compromise that prevents healing regardless of what else is done in the protocol.

Antibiotics present a different mechanism of gut disruption. Even a single course of antibiotics can produce significant alterations in gut microbial diversity that persist for six months to two years following treatment.[8] Multiple courses of antibiotics over years, which is not unusual in the history of patients with chronic illness and particularly those with recurrent sinus infections, urinary tract infections, or Lyme disease, can produce a degree of microbiome disruption that significantly impairs the ecosystem's capacity to maintain barrier integrity. The use of broad-spectrum antibiotics is not inherently problematic when clinically necessary, but the absence of microbiome restoration following antibiotic use is a significant missed opportunity in most medical settings.

Chronic Stress and the HPA-Gut Axis

The relationship between psychological and physiological stress and gut barrier function is mechanistically direct, not metaphorical. Corticotropin-releasing hormone (CRH), released from the hypothalamus in response to stress, activates mast cells in the gut wall through CRH receptor 1. Activated mast cells release histamine, proteases, and pro-inflammatory mediators that directly increase intestinal permeability and alter gut motility. This is a rapid effect: measurable changes in gut permeability have been documented in human studies within hours of acute psychological stress exposure.[9]

Chronic stress compounds this through cortisol dysregulation. Elevated cortisol impairs tight junction protein synthesis, suppresses secretory IgA production (the gut's primary immune defense), and alters the composition of the gut microbiome in ways that favor pathogenic over commensal organisms. The clinical consequence is a gut that is perpetually operating in a defensive, pro-inflammatory state. This is why patients who are under sustained stress rarely achieve full gut barrier repair even when their diet and supplement protocols are otherwise well-designed.

The bidirectionality of this relationship is equally important to understand. A compromised gut barrier generates chronic immune activation that drives neuroinflammation via inflammatory cytokine signaling to the brain. This neuroinflammation amplifies the stress response, increases anxiety, disrupts sleep architecture, and impairs the regulatory mechanisms that would normally allow the HPA axis to return to baseline after a stressor. Gut permeability and stress dysregulation feed each other in a cycle that requires both to be addressed for either to resolve.

Dietary Processed Foods, Seed Oils, and Emulsifiers

The industrial food supply contains several specific compounds that have been shown to impair gut barrier integrity through mechanisms independent of their general nutritional inadequacy. Emulsifiers, particularly carboxymethylcellulose (CMC) and polysorbate 80, which appear in a wide range of processed foods as texture agents, have been shown in multiple animal studies to disrupt the gut mucus layer, promote bacterial translocation across the epithelium, and drive low-grade intestinal inflammation.[10] Human data is more limited but consistent with the animal findings.

Refined seed oils high in linoleic acid, specifically canola, soybean, corn, sunflower, and safflower, present a different mechanism. The dramatic increase in dietary omega-6 fatty acids relative to omega-3 fatty acids that has accompanied the industrialization of the food supply drives the arachidonic acid cascade, generating pro-inflammatory prostaglandins and leukotrienes that degrade tight junction proteins and promote intestinal inflammation over time. The membrane phospholipids of intestinal epithelial cells incorporate the fatty acids available from the diet. A cell membrane built primarily from omega-6 fatty acids produces a fundamentally different inflammatory signaling profile than one built from omega-3s, and that signaling directly affects tight junction integrity.

Refined sugar and high-fructose corn syrup contribute to gut barrier compromise primarily through their effect on the microbiome. Pathogenic organisms, particularly Candida and certain dysbiotic bacteria, are highly preferential metabolizers of simple sugars. A diet high in refined carbohydrates selectively feeds the organisms most likely to damage the gut barrier while simultaneously depleting the fiber substrates that beneficial bacteria need to produce the short-chain fatty acids that maintain barrier integrity. The dietary change alone, removing refined sugar and processed foods, often produces measurable improvement in gut symptoms within two to three weeks, not because it directly repairs the barrier, but because it removes the primary fuel source for the organisms that are breaking it.

Environmental Toxins and the Mold Connection

The relationship between environmental toxin burden and gut barrier integrity has become increasingly well-documented in recent years. Glyphosate, the active ingredient in Roundup herbicide and now present in detectable levels in a wide range of commercially grown foods, has been shown to inhibit tight junction proteins and alter the gut microbiome in ways that favor pathogenic organisms. Several studies have documented measurable reductions in beneficial Lactobacillus and Bifidobacterium species in glyphosate-exposed gut microbiomes, with corresponding increases in pathogenic organisms.

Mycotoxins, the toxic secondary metabolites produced by mold species in water-damaged buildings, are directly toxic to intestinal epithelial cells and have been shown to suppress secretory IgA production, impair tight junction assembly, and promote gut barrier dysfunction. This is a central clinical reason why mold illness and leaky gut so frequently coexist. The patient who comes in with a history of water-damaged building exposure and presents with gut symptoms, fatigue, brain fog, and hormonal dysregulation is not experiencing two separate problems. The mold toxin burden is a primary driver of the gut barrier dysfunction that is producing the systemic symptom picture. In these patients, gut repair protocols that do not also address the underlying mycotoxin burden produce incomplete and often temporary results.

KEY 3

The Right Labs, Getting Answers That Actually Mean Something

Most patients with intestinal permeability have spent years getting tested. What they have not had is the right testing. There is a fundamental difference between the standard gastroenterological workup, which is designed to detect structural pathology, and the functional medicine approach to gut assessment, which is designed to characterize the gut's functional state: its barrier integrity, its microbial ecosystem, its immune competence, and the specific organisms and mechanisms driving dysfunction. The panels described below are what make accurate diagnosis and targeted treatment possible. But before any panel is described, one testing problem has to be stated plainly, because this book's credibility depends on it.

The Zonulin Testing Problem, Stated Honestly

Zonulin the molecule and zonulin the lab test are in very different shape, and conflating them is where both sides of this argument go wrong.

The biology is real. Zonulin (identical to pre-haptoglobin 2) is an established regulator of tight junction opening, and the research base behind it is the legitimate science this chapter began with.[1] The commercial testing is another matter: an analysis of the most widely used commercial zonulin ELISA found that the kit does not actually detect pre-haptoglobin 2, the molecule it claims to measure, and instead recognizes other proteins, including properdin, with measured "zonulin" concentrations failing to track the haptoglobin genotypes that determine whether a person can even produce zonulin.[15] In plain terms: the number on a commercial zonulin result, serum or stool, is not a validated measurement of the molecule it is named after. **This practice does not treat any zonulin level as a diagnosis, and neither should any practice.**

Skeptics then take one step too far in the other direction. A physician on social media telling you to "skip leaky gut testing because everyone has intestinal permeability" is right about the zonulin kits and wrong about the physiology. Of course everyone has intestinal permeability; you could not absorb nutrients without it. Dismissing intestinal hyperpermeability on those grounds is like dismissing hypertension because everyone has blood pressure. The clinical question is never whether the barrier is permeable but whether it has become excessively so, and mainstream gastroenterology treats that question as real: a review in the journal Gut lays out the intestinal barrier's components, the validated research measurements of permeability (orally administered probe molecules such as the lactulose-mannitol test, and tissue-based methods), the stressed states in which barrier function measurably deteriorates, and, in the same breath, the caution that public-domain "leaky gut" claims require confirmation before endorsing exclusion diets or repair supplements.[16] Both halves of that sentence are this book's position.

What this means practically for the panels below: the antibody and stool markers they carry are read as supportive pattern evidence alongside your symptoms, your history, and your response over time, never as stand-alone verdicts, and no single number on any of them, zonulin least of all, diagnoses a leaky gut by itself.

The Wheat Zoomer: Starting with Barrier Integrity

The Wheat Zoomer panel from Vibrant Wellness is the entry-level permeability assessment and the appropriate starting point for most patients presenting with gut-related symptoms. Its value is in what it measures directly: the antibody response to the tight junction proteins themselves. Occludin/zonulin antibodies are interpreted as evidence that the immune system has encountered these proteins, which is expected mainly when they have reached systemic circulation from a compromised epithelium. Actomyosin IgA and IgG antibodies reflect intestinal epithelial cell injury, and the actin-antibody family is the best-validated marker in this class, though that validation is strongest specifically in celiac disease with villous atrophy rather than for permeability in general. The honest grade for the panel as a whole: suggestive immunological pattern evidence, meaningfully better than a bare zonulin number, and still not a validated stand-alone diagnostic for barrier compromise. It is read alongside the clinical picture, per the testing-honesty section above, and the companion lab guide in the Gut kit walks through each marker's actual validation study by study.

The Wheat Zoomer also measures antibody reactivity to 26 different wheat and gluten peptides, providing a far more sensitive picture of gluten reactivity than the standard celiac panel. Most patients who have had standard celiac antibody testing and received a negative result have not had the breadth of gluten peptide testing that the Wheat Zoomer provides. Non-celiac gluten sensitivity, a real and measurable immunological phenomenon in which the immune system reacts to gluten peptides without producing the villous atrophy of celiac disease,[11] is identified on the Wheat Zoomer in patients who test negative on standard celiac panels. The clinical significance for gut repair is substantial: a patient cannot repair a gut barrier that gliadin is continuing to disrupt.

The Wheat Zoomer is available at $479 retail through rootcauseunlocked.com/order-labs and is the recommended first-line panel for patients with gut-related symptoms, food sensitivities, skin conditions, hormonal dysregulation, or brain fog without a known autoimmune diagnosis.

The Gut Zoomer Complete: Mapping the Ecosystem

For patients with significant dysbiosis symptoms, including persistent bloating and gas, alternating bowel habits, suspected parasites, history of H. pylori, history of Lyme disease or chronic infection, or prior antibiotic use, the Gut Zoomer Complete from Vibrant Wellness provides the comprehensive microbial picture needed to make treatment precise rather than speculative. The panel characterizes over 300 gut organisms using DNA sequencing technology, identifying not just the presence of pathogenic organisms but the relative abundance and diversity of the entire microbial ecosystem.

The Gut Zoomer measures fecal zonulin (which carries the same assay-validity problem described in the testing-honesty section above, and is read here as a minor supporting data point rather than a direct permeability measure), fecal gliadin (measuring active gluten exposure in the gut lumen), secretory IgA, the gut's primary immune defense. When depleted, it indicates chronic immune suppression and predicts susceptibility to ongoing pathogenic colonization), and short-chain fatty acid production, which reflects the functional capacity of the microbiome to support the colonocyte fuel supply. It identifies specific pathogenic organisms including H. pylori (with virulence factor assessment), Candida species, Giardia, Cryptosporidium, Blastocystis hominis, and a range of dysbiotic bacterial overgrowths. At $780 retail, it provides a degree of clinical specificity that makes every subsequent treatment decision more targeted and more effective.

The GI-MAP: A Lower-Cost Alternative

For patients who need a lower-cost alternative to the Gut Zoomer Complete, the GI-MAP from Diagnostic Solutions is a solid option at $521 retail. It uses quantitative PCR technology to detect specific gut pathogens with a high degree of sensitivity and is particularly useful for identifying H. pylori, Candida, parasites, and bacterial overgrowth. It is worth noting that the GI-MAP does not measure intestinal permeability markers directly, does not assess secretory IgA or short-chain fatty acid production, and does not provide the full ecosystem picture of the Gut Zoomer Complete. For patients whose primary concern is pathogen identification rather than comprehensive barrier and microbiome assessment, it is a clinically useful starting point. The GI-MAP is ordered through the office.

The Organic Acids Test: When the Brain Is Involved

When gut symptoms are accompanied by significant brain fog, mood instability, fatigue that is disproportionate to sleep quantity, or cognitive symptoms, the Organic Acids Test (OAT) adds a dimension to the picture that gut-specific panels cannot provide. The OAT measures urinary metabolites, specifically the byproducts of cellular metabolism, microbial activity, and neurotransmitter processing, that reflect the biochemical environment in which the gut and brain are operating.

The OAT's fungal overgrowth markers, including citrinin and various Aspergillus metabolites, can identify fungal burden that is not detectable on standard stool panels because the organisms are colonizing the small intestine rather than the colon. Its neurotransmitter metabolite markers, which measure the downstream products of dopamine, serotonin, and GABA metabolism, reveal whether the neurochemical consequences of gut dysfunction are already manifesting in the nervous system. For patients whose primary complaint is fatigue and cognitive impairment alongside gut symptoms, the OAT often provides the most clinically actionable findings of any single panel. It is available at $571 retail.

Blood Chemistry: The Systemic Inflammatory Baseline

Before or alongside any specialty gut testing, a comprehensive blood chemistry panel establishes the systemic inflammatory and metabolic baseline that contextualizes the gut findings. High-sensitivity CRP reflects the degree of systemic inflammatory burden. CBC with differential may reveal eosinophilia suggesting parasitic infection or significant food reactivity. The comprehensive metabolic panel assesses liver function, which is the primary detoxification organ bearing the burden of endotoxin clearance from a leaking gut. Thyroid markers, fasting glucose, and insulin provide the hormonal and metabolic picture that determines how broadly the gut dysfunction has affected downstream systems. The blood chemistry panel runs at $318 retail through Professional Co-op.

Testing Decision Framework

Gut symptoms + food sensitivities + no known autoimmunity: start with Wheat Zoomer ($479)

Significant bloating, altered bowels, suspected H. pylori or parasites: Gut Zoomer Complete ($780)

Lower-cost alternative to Gut Zoomer Complete: GI-MAP through the office ($521)

Brain fog, mood changes, or fatigue alongside gut symptoms: add Organic Acids Test ($571)

Confirmed or suspected Hashimoto's: add Yersinia serology and see Key 7

All panels available at rootcauseunlocked.com/order-labs

KEY 4

Drainage First, The Step That Makes Everything Else Work

If there is one concept in this entire guide that separates a protocol that produces lasting results from one that produces temporary improvement or makes the patient feel worse before they feel better, it is this: the drainage pathways must be open before the gut repair protocol begins. This step cannot be abbreviated or skipped. It is the clinical foundation on which every other intervention depends.

When the gut repair process is initiated, when pathogens begin dying, when the barrier starts to release accumulated debris, when the microbiome shifts and produces changes in microbial metabolite output, large quantities of cellular waste, endotoxins, and inflammatory byproducts enter the lymphatic system and systemic circulation. The body's ability to process and eliminate these substances depends on three drainage systems functioning adequately: the lymphatic system, which clears interstitial fluid and cellular waste; the liver, which conjugates and neutralizes toxins for biliary elimination; and the kidneys, which clear water-soluble metabolic waste products.

In patients with chronic gut dysfunction, all three of these drainage systems are typically under stress. The liver has been processing elevated LPS endotoxin loads for months or years. The lymphatic system is congested from chronic inflammation. Methylation, the biochemical process that drives Phase 2 liver detoxification and is essential for clearing endotoxins, hormones, neurotransmitter metabolites, and environmental toxins, is frequently impaired. When gut repair is initiated without first restoring drainage capacity, the waste products that are mobilized have nowhere to go. They recirculate, redeposit in tissues, and produce what is clinically called a Herxheimer reaction: a worsening of symptoms that patients often interpret as a sign that the protocol is wrong, when in fact it is a sign that drainage was not prepared before the repair process was started.

The Drainage Foundation

At Root Health, every gut repair protocol begins with a minimum of two weeks of drainage support before any targeted repair agents are introduced. The core drainage triad consists of three products that address the three primary drainage bottlenecks.

Lymphatic and extracellular matrix support is the piece most relevant for the majority of gut dysfunction patients. The extracellular matrix, the fluid-filled connective tissue space through which cellular waste must pass on its way to the lymphatic vessels, becomes congested in patients with chronic inflammation, dietary toxin burden, and xenobiotic accumulation. A drainage botanical formula aimed at this pattern of matrix congestion, typically taken twice daily in warm water, is the appropriate choice for most patients presenting with gut dysfunction and a history of dietary toxin exposure, processed food consumption, or chemical sensitivity. Adequate water intake and daily bowel movements do more of this work than any bottle, and they cost nothing.

The drainage support has to match the stage. Acute formulas, generally built around lymphagogue botanicals such as red root, cleavers, and echinacea, suit recent inflammation and active infection. Deeper, longer-acting blends suit longstanding degeneration, heavy metal burden, and conditions present for more than three years. These are not stacked on top of each other; each addresses a distinct level of drainage compromise, and the selection depends on chronicity, severity, and the nature of the burden. For the majority of gut repair cases, the middle option aimed at matrix congestion is the right one.

Methylated B12 addresses the methylation capacity required for Phase 2 liver detoxification. The liver's ability to conjugate and neutralize the endotoxins, hormones, and environmental toxins generated during gut repair depends on an adequate supply of methyl groups. Many patients with chronic gut dysfunction have compromised methylation capacity due to nutrient depletion, genetic variants in methylation enzymes (particularly MTHFR), or the ongoing metabolic burden of processing a chronically leaking gut. A liposomal or sublingual format is preferable to an oral tablet here, because it bypasses the gut entirely, which is clinically relevant in patients whose gut is too compromised to reliably absorb what they swallow. Look for methylcobalamin rather than cyanocobalamin.

A full-spectrum electrolyte and trace mineral supplement provides the foundation required for cellular hydration and efficient drainage at the cellular level. Dehydration at the intracellular level, which can exist in patients who drink adequate water but have compromised mineral balance, is a bottleneck in drainage capacity that turns up with unusual consistency. Without adequate cellular hydration, the intracellular fluid dynamics that drive waste products from cells into the interstitial space and ultimately into lymphatic vessels are impaired. an electrolyte and trace mineral base is dosed per label direction and is maintained throughout the full repair protocol.

Two weeks on the drainage triad before introducing any gut repair agents is the minimum. In patients with significant chronicity, confirmed heavy metal burden, or a history of multiple antibiotic courses, extending the drainage phase to three or four weeks often produces substantially better outcomes in the repair phases that follow.

The Anti-Inflammatory Gut Repair Diet

Dietary change is the ground on which gut repair either succeeds or fails, not a supplementary measure. The most precisely targeted supplement protocol in the world will not produce lasting gut barrier restoration in a patient who continues to consume the foods and substances that are breaking the barrier down daily. This is basic physiology, not a matter of willpower or lifestyle preference. You cannot repair a structure that is being continuously damaged.

The dietary approach in this section is organized in phases because the clinical rationale for each phase is different, and because asking a patient to make every change simultaneously increases the likelihood that none of the changes will be sustained. The sequence matters, and understanding why each phase is necessary makes compliance far more likely than a list of rules to follow.

Phase One: The Non-Negotiables

The first phase of dietary intervention is defined by removal, specifically the removal of the dietary factors with the most direct and documented effects on gut barrier integrity. These are foods and substances that actively disrupt the tight junction architecture or feed the pathogenic organisms that are damaging it, not foods that are merely suboptimal.

Gluten and all wheat-containing foods must be completely eliminated, not reduced, during the active repair phase. The research basis is that gliadin activates the zonulin pathway in intestinal tissue regardless of celiac status,[3] so even small and intermittent gliadin exposures can re-trigger the signaling this phase is trying to quiet. Patients who attempt a “mostly gluten-free” approach do not achieve the same degree of barrier restoration as those who eliminate it completely. Cross-contamination matters. For patients who test positive for gluten reactivity on the Wheat Zoomer, eating at restaurants that use shared cooking surfaces with gluten-containing foods can be sufficient to maintain a state of chronic tight junction disruption. During the active repair phase, strict elimination is the clinical standard.

Conventional dairy, particularly casein A1 protein from most commercially available cow’s milk products, presents a mechanism similar to gliadin in genetically susceptible individuals. A1 casein is cleaved during digestion to produce beta-casomorphin-7, a peptide that has been shown to increase intestinal permeability through mechanisms that involve both direct epithelial effects and opiate receptor-mediated changes in gut motility and secretion. Some patients tolerate A2 dairy, from certain breeds of cows, goats, and sheep, without the same permeability consequences, and this can be tested during the reintroduction phase. During active repair, complete dairy elimination is the more conservative and clinically appropriate approach.

Refined sugar and high-fructose corn syrup need to be understood not as empty calories but as a specific ecological intervention in the gut microbiome. Candida and dysbiotic bacteria are highly preferential metabolizers of simple sugars. Continuing to consume refined sugar during a gut repair protocol while simultaneously attempting to reduce Candida overgrowth with supplementation is clinically counterproductive: it is the equivalent of trying to drain a bathtub while the tap is running. For patients with confirmed or suspected Candida overgrowth or significant dysbiosis, removing refined sugar is prerequisite, not optional.

Seed oils, specifically canola, soybean, corn, sunflower, safflower, and cottonseed oils, need to be replaced, not merely reduced, because their effects on gut barrier integrity operate through the composition of the cell membranes themselves. The linoleic acid in seed oils is incorporated into the phospholipid bilayers of intestinal epithelial cells, and cell membranes built primarily from omega-6 fatty acids produce a fundamentally different inflammatory signaling profile than those built from omega-3s and saturated fats. This is not a change that produces results in days, as membrane phospholipid turnover takes weeks to months, but it is a change that affects the structural integrity of every cell in the gut lining and cannot be bypassed.

Alcohol, even in moderate quantities, directly increases intestinal permeability through acetaldehyde, a toxic metabolite of alcohol metabolism that is particularly damaging to tight junction proteins, and through direct suppression of secretory IgA production. There is no safe level of alcohol consumption during an active gut repair protocol. The literature is unambiguous on this point. In clinical practice, the most common reason a patient's gut repair protocol stalls after initial improvement is continued alcohol consumption, even at levels that would conventionally be described as moderate or social.

Phase Two: What to Add

After the barrier-disrupting foods are removed, the second phase focuses on foods that actively support the repair process. This is where the nutritional quality of the diet shifts from merely avoiding harm to providing the specific substrates that the gut lining needs to regenerate.

Bone broth deserves particular attention because its therapeutic value in gut repair rests less on its mineral content than on its amino acid composition. The slow simmering of collagen-rich bones and connective tissue produces a broth concentrated in glycine, proline, and hydroxyproline: the three amino acids that are the primary structural constituents of the collagen matrix underlying the intestinal epithelium and of the tight junction proteins themselves. Glycine, in particular, has anti-inflammatory effects at the level of the gut mucosa that extend beyond its role as a collagen precursor. Daily consumption of bone broth during the active repair phase provides a sustained supply of the raw materials the gut lining needs to rebuild.

Cooked vegetables, particularly during the early repair phase, are preferable to raw for most patients with significant gut inflammation. The fiber in raw vegetables, while ultimately beneficial for microbiome diversity, requires substantial digestive work and can be irritating to an inflamed gut lining in the short term. Cooking softens the fiber matrix and makes nutrients more bioavailable. Zucchini, sweet potato, beets, carrots, winter squash, and well-cooked leafy greens provide a range of polyphenols, prebiotic fibers, and micronutrients that support both barrier repair and the reestablishment of a diverse microbiome without the irritation risk of raw fibrous vegetables.

Fermented foods warrant a clinical caveat that is frequently overlooked in generic gut health advice. Sauerkraut, kimchi, kombucha, kefir, and other fermented foods are excellent sources of diverse probiotic organisms and are genuinely supportive of microbiome recovery in patients with intact histamine clearance capacity. However, fermented foods are among the highest-histamine foods in the human diet. In patients with concurrent histamine intolerance, which is extremely common in leaky gut cases because gut permeability and DAO enzyme dysfunction frequently coexist, consuming fermented foods during the repair phase can provoke flushing, headaches, hives, heart palpitations, anxiety, and gut symptom worsening that is often misattributed to the fermented food being “detoxing” rather than recognized as a histamine reaction. The appropriate approach is to assess for histamine intolerance before introducing fermented foods rather than assuming they will be universally beneficial.

Phase Three: Systematic Reintroduction

After 30 to 60 days of Phase One and Two implementation, tolerance to removed foods can be systematically assessed. The protocol is straightforward: introduce one food at a time, every 72 hours, and document symptom responses across the full 72-hour window. The 72-hour window is important because IgG-mediated food sensitivity reactions, the type most commonly associated with intestinal permeability, are delayed, typically appearing 24 to 72 hours after exposure rather than immediately. Patients who test single foods over 24-hour windows frequently miss the delayed reactions that are most clinically relevant.

The reintroduction process is as much a diagnostic tool as a recovery milestone. Foods that provoke reactions during reintroduction, even subtle ones such as mild brain fog, slight skin changes, increased bowel frequency, and energy changes, reveal sensitivities that had been masked by global gut inflammation during the elimination phase. These findings directly inform the long-term dietary strategy and often explain why patients with leaky gut had developed apparently paradoxical sensitivities to foods they had always previously tolerated.

KEY 6

Strategic Supplementation for Gut Repair

The supplement protocol for gut repair is organized as a sequence, not a stack. The distinction matters clinically. A stack is a collection of supplements taken simultaneously without regard for the order of physiological events. A sequence reflects the understanding that the gut's recovery process has phases, and that introducing certain interventions before others are in place either reduces their efficacy or produces adverse reactions. The sequence described here, drainage, barrier repair, microbiome restoration, and targeted pathogen support, follows the mechanistic order in which these interventions must occur to produce the outcomes patients are looking for.

The clinical protocols used at Root Health draw primarily from practitioner-grade supplement lines that are not available over the counter and are dispensed through the practice. These formulations are used because their delivery technology, potency, and ingredient sourcing meet the standards required for clinical outcomes. Over-the-counter alternatives exist for many of the categories described below, including amino acid gut repair formulas, probiotics, and antimicrobial botanicals. Those options are reviewed and individualized as part of the Root Restore program and the supplement guide provided to enrolled patients. The framework here explains the clinical reasoning behind each phase so that you understand what the protocol is doing and why, regardless of which specific products are ultimately used in your case.

Phase One: Drainage (Weeks 1 through 2, continued throughout)

The drainage foundation, specifically a lymphatic botanical, methylated B12, and an electrolyte and trace mineral base, is introduced in the first two weeks and maintained throughout the full protocol. As described in Key 4, these are the prerequisite interventions that determine whether everything that follows works. No other supplements are introduced until the drainage triad has been running for a minimum of two weeks.

Phase Two: Gut Barrier Repair (Weeks 3 through 6)

L-glutamine is the primary gut barrier repair agent at Root Health and the most clinically important supplement in this phase. L-Glutamine is the preferred fuel source for intestinal enterocytes, the cells that line the gut wall, and is the primary amino acid substrate for tight junction protein synthesis. The research on L-Glutamine and gut barrier repair is extensive: it increases tight junction expression, reduces gut permeability markers, supports secretory IgA production,[12] and provides the nitrogen substrate for the rapid cell turnover that characterizes the intestinal epithelium, which replaces itself completely every three to five days. Look for a powder that combines glutamine with proline and glycine, the amino acids that support the collagen matrix underlying the epithelium. It is typically dosed at 5 to 10 g twice daily on an empty stomach, which maximizes availability to the gut epithelium before systemic tissues metabolize it.

Bovine colostrum addresses a component of gut repair that is frequently absent from conventional gut healing protocols: the restoration of gut-associated lymphoid tissue function. The GALT houses the immune infrastructure that maintains the gut's immunological relationship with its microbial inhabitants, distinguishing between commensal organisms that should be tolerated and pathogens that should be resisted. In patients with chronic gut dysfunction, prolonged NSAID use, or history of immunosuppressive medications, GALT function is often significantly impaired. Without restoring the GALT's immunological competence, the microbiome may not stabilize even after the barrier has been physically repaired, because the immune surveillance mechanism that should be regulating microbial ecology is not functioning.

A bioavailable curcumin provides upstream anti-inflammatory support at the NF-kB and COX-2 level, reducing the inflammatory signaling that impairs tight junction protein expression. Delivery format is the critical distinction here, because standard curcumin is notoriously poorly absorbed and most studies showing clinical benefit used either high doses or an enhanced delivery format such as a phytosome, nanoparticle, or piperine-paired preparation. Buying plain curcumin powder and expecting the trial results is the common mistake.

Beyond the core protocol, there is a category of gut barrier support nutrients that address the structural components of the barrier directly: compounds that fortify tight junction proteins, support the mucus layer overlying the epithelium, and provide the glycosaminoglycan substrates needed for glycocalyx integrity. These include zinc carnosine, deglycyrrhized licorice, N-acetyl glucosamine, and colostrum, among others. The specific formulations and dosing for these compounds are part of the personalized supplement guide provided to enrolled patients, because the appropriate selection depends on lab findings, symptom pattern, and the presence or absence of concurrent H. pylori, NSAID history, or active gastric inflammation.

Phase Three: Microbiome Restoration (Weeks 4 through 8)

A spore-based probiotic (Bacillus subtilis, Bacillus coagulans, and related species) is introduced in the fourth week, once barrier repair has been established for at least two weeks. The spore format confers a practical advantage over conventional preparations: spore-forming organisms survive gastric acid transit reliably and germinate in the small intestine, where conventional lactobacillus and bifidobacterium preparations frequently do not arrive intact. The GALT-stimulating properties of these strains are particularly relevant where secretory IgA depletion has been a significant feature of the picture.

When confirmed dysbiosis on testing indicates a more substantial microbiome reconstruction need, a high-dose multi-strain probiotic in the range of 50 billion CFU accelerates the reseeding process. This product is reserved for cases where lab findings confirm significant microbial diversity depletion. Introducing high-dose probiotics into a gut with significant ongoing inflammation and an unrepaired barrier can paradoxically worsen symptoms by increasing the microbial antigen load before the immune system is prepared to handle it.

Saccharomyces boulardii serves a specific and important role that is distinct from bacterial probiotic preparations. S. boulardii is a beneficial yeast that does not permanently colonize the gut but produces transient clinical effects that are highly valuable during the repair phase: it competes directly with Candida for mucosal binding sites, it stimulates secretory IgA production, it produces proteases that degrade Clostridium difficile toxins, and it stabilizes the gut barrier through effects on tight junction protein expression. For any patient with confirmed or suspected Candida overgrowth, which should be considered in any patient with a history of antibiotic use, sugar-heavy diet, or positive OAT fungal markers, S. boulardii is a standard component of the microbiome restoration phase.

Phase Four: Targeted Pathogen Support (Weeks 6 through 12)

Targeted antimicrobial and pathogen-specific interventions are the last category to be introduced, not the first. This sequence is counterintuitive for many patients, who assume that identifying a pathogen on a lab result means addressing the pathogen is the highest priority. The clinical reality is that antimicrobial interventions in a gut with unrepaired barrier function and insufficient drainage capacity produce excessive die-off reactions, systemic toxin burden, and often a rebound in the very pathogen populations they are targeting, because the competitive microbial community that would normally keep them in check has not been restored.

Once drainage is established and the barrier has been supported for at least two weeks, pathogen-specific interventions are selected based on lab findings. For Candida overgrowth, pau d'arco and enteric-coated oregano oil standardized for carvacrol form the botanical antifungal core, and the full sequencing is covered in the companion Candida material. For H. pylori, mastic gum, and for general bacterial overgrowth and dysbiosis, berberine and a broad-spectrum botanical antimicrobial, are the usual choices. For Yersinia enterocolitica, the foodborne pathogen with documented thyroid autoimmune consequences, berberine has shown in vitro activity, and it is combined with gut barrier repair to reduce the antigen traffic driving molecular mimicry. Note that in vitro activity is a laboratory finding rather than a demonstrated clinical outcome, and that confirmed H. pylori is a condition where prescription eradication therapy has the evidence behind it.

Phase	Products	Duration	Clinical Goal
1. Drainage	Lymphatic botanical blend, methylated B12, electrolyte and trace mineral base	Weeks 1-2 min; continue throughout	Open lymphatic, liver Phase 2, cellular hydration
2. Barrier Repair	L-glutamine with glycine and proline, colostrum, bioavailable curcumin, plus targeted barrier nutrients such as zinc carnosine and deglycyrrhizinated licorice	Weeks 3-6	Rebuild tight junctions, restore mucosal immunity
3. Microbiome	Spore-based probiotic, high-dose multi-strain probiotic if indicated, Saccharomyces boulardii	Weeks 4-8	Reseed diversity, restore secretory IgA, displace Candida
4. Pathogen	Selected by lab findings: pau d'arco, enteric-coated oregano oil, mastic gum, berberine	Weeks 6-12	Targeted elimination with drainage and barrier in place

Where to get these

Every item in this table is a single ingredient or a standardized botanical available from any reputable retailer. Look for independent third-party testing and match the form named here rather than the name on the front of the bottle. If you would rather not evaluate that yourself, Root Health keeps a curated dispensary organized by these phases.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. Nothing in this book was included, excluded, ranked, or dosed based on what it earns, and you can buy any of it anywhere.

KEY 7

When the Gut Attacks the Thyroid

Leaky gut does not stay local. In some patients, a compromised gut barrier goes beyond contributing to systemic symptoms. It is the hidden driver of a specific autoimmune process targeting a completely different organ: the thyroid gland. Two infectious organisms that originate in the gut have well-documented mechanisms by which they trigger or amplify Hashimoto's thyroiditis. Understanding these connections may be the most important clinical insight in this entire guide for patients who have both gut symptoms and thyroid problems that are not responding to conventional treatment.

Yersinia Enterocolitica and the Thyroid

Yersinia enterocolitica is a gram-negative foodborne bacterium transmitted primarily through contaminated pork products, unpasteurized dairy, and undercooked meats. In immunocompetent individuals, acute Yersinia infection produces a self-limiting gastroenteritis that resolves within a few weeks. What is not widely appreciated is that Yersinia can persist in the gut in a subclinical state in susceptible individuals, and in doing so it triggers a molecular mechanism with thyroid consequences that conventional medicine rarely considers.

Yersinia produces an outer membrane protein that shares structural homology with the human TSH receptor on thyroid cells. When the immune system generates antibodies against this Yersinia surface protein, which is appropriate and necessary, those antibodies cross-react with thyroid tissue through a mechanism called molecular mimicry.[13] The immune system, unable to distinguish between the bacterial epitope and the thyroid receptor it resembles, mounts an autoimmune attack on the thyroid gland. The result is the elevated TPO and thyroglobulin antibodies characteristic of Hashimoto's thyroiditis. Multiple studies have documented significantly elevated Yersinia antibodies in Hashimoto's patients compared to healthy controls, and the connection between foodborne Yersinia exposure and thyroid autoimmunity is now well-established in the immunological literature.[14]

The clinical implication is important: Yersinia is a gut pathogen with a thyroid consequence. It is not covered on standard thyroid panels. It is not a tick-borne organism and is not covered on tick-borne disease panels. It requires a standalone order: LabCorp Yersinia enterocolitica IgG, IgA, and IgM serology. Treating Hashimoto's without assessing for Yersinia is treating the downstream autoimmune consequence while leaving a primary upstream trigger unaddressed.

Lyme Disease and the Thyroid

Borrelia burgdorferi, the primary causative organism in Lyme disease, has an independent and clinically significant relationship with Hashimoto's thyroiditis that goes beyond the general immunological dysfunction of chronic Lyme. Borrelia produces surface proteins with structural similarity to thyroid antigens, triggering cross-reactive immune responses through the same molecular mimicry mechanism as Yersinia.[13] Chronic

Lyme disease also drives a sustained Th17-dominant immune environment, the same immune polarization that characterizes the lymphocytic thyroid infiltration of Hashimoto's, making the two conditions mutually reinforcing in patients who have both.

The gut is the shared mechanism that ties these two conditions together. Lyme disease suppresses secretory IgA production, promotes gut dysbiosis, and compromises gut barrier integrity. A leaky gut in a patient with active or past Lyme infection creates the persistent antigen trafficking that keeps both the Lyme immune response and the thyroid autoimmunity running simultaneously. In clinical practice, patients with Hashimoto's whose antibody levels do not normalize despite appropriate dietary intervention and gut repair should be evaluated for tick-borne co-infection, because the immune dysregulation driving their antibody elevation may have an infectious origin that gut repair alone cannot resolve.

The Key Point for Hashimoto's Patients

Both Yersinia and Lyme are gut-mediated triggers for Hashimoto's thyroiditis. Treating the gut is a prerequisite for thyroid autoimmune recovery, but in some patients, the gut alone is insufficient. The full testing and treatment protocols for both connections, including the Yersinia serology sequence, the Lyme co-infection workup, and the complete Hashimoto's supplement protocol, are in the companion guide:

Unlock Your Hashimoto's · rootcauseunlocked.com/hashimotos

KEY 8

Your Next Steps with Root Health

The framework in this guide gives you something that most patients with gut-related chronic illness have never had: a mechanistic understanding of what is actually happening in their body, why the standard workup misses it, and what a systematic approach to recovery actually looks like. That understanding is clinically valuable on its own. But it reaches its full potential when it is applied to your specific case, with your labs, your history, and your specific pattern of dysfunction.

Gut dysfunction is not a one-size-fits-all condition. The specific organisms driving your dysbiosis, the degree of your barrier compromise, the genetic variants affecting your detoxification capacity, the co-conditions, including thyroid autoimmunity, tick-borne infection, mold burden, and hormonal dysregulation, all of which interact with your gut picture all require clinical interpretation. The protocols in this guide are the clinical framework. Individualized care is what makes them work for you specifically.

Start with the Right Testing

Your entry lab panel should match your clinical picture. For patients with gut symptoms, food sensitivities, and no known autoimmunity, the Wheat Zoomer at $479 is the right starting point. It directly measures barrier integrity and gluten reactivity. For patients with significant bloating, altered bowels, or suspected pathogenic overgrowth, the Gut Zoomer Complete at $780 provides the most comprehensive microbial picture. If cost is a primary consideration, the GI-MAP at $521, available through the office, is a lower-cost alternative. When brain fog, anxiety, or fatigue are prominent alongside gut symptoms, the Organic Acids Test at $571 adds the neurochemical dimension. For patients with confirmed or suspected Hashimoto's, Yersinia serology belongs in the workup. All panels are available at rootcauseunlocked.com/order-labs.

Book a Root Discovery Session

A Root Discovery Session is a complimentary 30-minute clinical conversation with Dr. Seay. It is not a sales call. It is a focused clinical intake where your symptom history, prior testing, and current clinical picture are reviewed to identify the most targeted path forward for your case. If you are uncertain which lab panel is right for you, or if you want clinical input on interpreting prior results before investing in additional testing, this is where that conversation happens. Book at rootcauseunlocked.com/discovery.

The Root Restore Program

For patients who are ready for structured, supervised 1-on-1 clinical care, the Root Restore Program is a 6-month protocol that includes comprehensive testing, a personalized drainage-first supplement protocol, dietary guidance, and ongoing clinical oversight at every phase of the recovery process. Payment plans are available. Learn more at rootcauseunlocked.com/gut.

A Note from Dr. Seay

If you have been suffering with gut symptoms, fatigue, brain fog, skin conditions, or hormonal disruption that nobody has been able to explain: please know that there is almost always a root cause that has not yet been found. A normal colonoscopy is not a complete gut workup. Normal bloodwork is not normal function. You deserve the right tests with the right clinical interpretation, and a protocol designed around your specific case. That is exactly what Root Health is built for.

Dr. Adam Seay, DC | Root Health, Garner NC | rootcauseunlocked.com

Root Health | Garner, NC | rootcauseunlocked.com/gut

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